

RECORD GROUPING AND RECONSTRUCTION FROM REPEATED PRIVATE AGGREGATES

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ABSTRACT

Private aggregate releases with the same record-level differential privacy guarantee and the same distribution of their average can still permit different reconstruction accuracy when the same records are grouped differently across rounds. We show that this difference arises only when the other records are unknown: records shared across rounds make the uncertainty structured rather than independent. At the same total guarantee, repeating one aggregate adds no information; what matters is whether changing participation reveals the unknown records. We characterize when averaging all releases is the best unbiased linear estimate, namely when rounds have equal total reuse, and we compare repeated releases with a one-time private mean, showing that records present in every round do not average out. We further show that rounds without the target can improve reconstruction despite its stronger guarantee. Learned attacks on bounded data show the same orderings.

Index Terms— Differential privacy, reconstruction, repeated releases, record grouping, generalized least squares

1. INTRODUCTION

Repeated releases of noisy aggregates can have identical record-wise differential privacy (DP) calibration [1] but reveal different amounts of information about a record. When the other records are unknown, their overlap across releases creates dependence that an attacker may exploit for reconstruction. We study this effect in a fixed-query model, separating how often records are used from how their uses are arranged across rounds. We characterize when the sequence of noisy batch sums, the *transcript*, improves on its average, and use a one-time private mean, the *source*, as a reference.

Fixing the batch sizes, how often each record is used, and the noise levels fixes each record’s privacy guarantee and, for identically distributed records, the distribution of the transcript average. Can the same records, grouped differently across rounds, still lead to different reconstruction errors? The answer depends on what the attacker knows [2]. If all other records are known, their values can be subtracted; under our calibration, when the target appears in every round, the

transcript and the source then contain the same information about it (Section 2). If the other records are unknown, however, each one contributes the same unknown value whenever it appears, so the resulting uncertainty is shared across rounds rather than independent. Comparing releases can then reveal part of this shared contribution and improve reconstruction.

Under the same total privacy guarantee, repeating one batch in every round adds nothing, for any record distribution; what matters is whether changing participation reveals the unknown records. First, with the reuse counts held fixed, averaging all releases is the best unbiased linear estimator exactly when the records in every round have the same total reuse count; we call such schedules *round-balanced*. For Gaussian non-target records, these schedules also reveal the least information among schedules with the same reuse counts. Second, relative to a one-time private mean, records used in only one round average out, while records used in every round remain, giving an explicit threshold between the two. Third, if some rounds omit the target, those rounds can help estimate the other records and improve reconstruction of the target, even though the target has a stronger privacy guarantee.

Related work. Prior work asks when aggregate queries allow reconstruction [3, 4]. Bayesian views of DP [5] and reconstruction bounds [2, 6] study how prior knowledge affects privacy, and uncertainty about the other records can itself protect a record [7, 8]. Secure aggregation hides individual updates [9], but repeated aggregates can still reveal individual contributions [10]; choosing which users appear together can limit this leakage without added noise [11]. Work on private linear queries studies query geometry [12], workload design [13], and Gaussian mechanisms [14]. We instead fix the privacy guarantee and ask what the full sequence reveals beyond its average.

2. OBSERVATION AND PRIVACY MODEL

We study reconstruction of a scalar target record X_* , and write $X_* = x$ when conditioning on its value. The attacker knows the population distribution and the public participation schedule, but not the realized non-target values. The schedule is fixed independently of the records.

Repeated transcript. We first assume that the target appears in every batch, each of size $B \geq 1$. The schedule $A \in \{0, 1\}^{T \times M}$ encodes the participation of M distinct non-target records over $T \geq 1$ rounds, with $A_{ti} = 1$ when record i appears in round t . Each row therefore has $B - 1$ ones. Each query is a batch sum or a known affine transformation with nonzero slope. Inverting this transformation and subtracting the expected non-target contributions gives

$$Y = x\mathbf{1}_T + Au + \eta, \quad (1)$$

where u contains the centered non-target values and η is the mechanism noise. The same u_i appears whenever record i is used. Section 3.4 covers partial target participation.

One-time benchmark. The *source* is a one-time noisy mean of $n \geq 1$ records, including the target. Multiplying the release by n and subtracting the expected non-target sum gives

$$Z_s = x + \sum_{i=1}^{n-1} u_i^s + \eta_s,$$

where u_i^s is a centered non-target value and η_s is the rescaled mechanism noise.

Privacy calibration. We calibrate both mechanisms on the bounded input domain $[-B_x, B_x]$, where $B_x > 0$, under replace-one adjacency. A Gaussian mechanism with sensitivity Δ and noise standard deviation Δ/μ , where $\mu > 0$, satisfies μ -Gaussian differential privacy (GDP). Squared GDP parameters add under independent composition [15]. Using exact sensitivities, we assign μ to the source and $\mu/\sqrt{T}$ to each transcript round. With $\nu_\mu := 4B_x^2/\mu^2$, the noises satisfy

$$\eta_s \sim \mathcal{N}(0, \nu_\mu), \quad \eta \sim \mathcal{N}(0, T\nu_\mu I_T). \quad (2)$$

The mechanism noises are independent of all records. Because the schedule is fixed and public, there is no subsampling amplification. Under full target participation, the target has parameter μ in both mechanisms. A non-target record used in $d_i := \sum_t A_{ti}$ rounds has parameter $\mu\sqrt{d_i/T} \leq \mu$, so every record satisfies the common bound μ . When comparing schedules, we keep the batch sizes, reuse counts, and mechanism noise fixed. Comparisons with the source share only the common record-level bound.

Assumption 1. For each mechanism, the distinct centered non-target values are mutually independent, have common variance $r_x > 0$, and are independent of $X_\star$.

Under Assumption 1, the source variance is $v_s := \text{Var}(Z_s | x) = \nu_\mu + (n-1)r_x$, and the transcript covariance is $\Sigma_A := \text{Cov}(Y | x) = r_x G_A + T\nu_\mu I_T$. Here $G_A := AA^\top$, and $(G_A)_{ts}$ counts the records shared by rounds t and s .

Structured uncertainty. If the non-target values are known, they can be subtracted from both observations. The source then reduces to $x + \eta_s$, while the transcript reduces to $x\mathbf{1}_T + \eta$. Its average, $x + \bar{\eta}$, is sufficient, and both observations

are equivalent to x plus $\mathcal{N}(0, \nu_\mu)$ noise, regardless of the schedule. When the non-target values are unknown, they instead contribute the covariance $r_x G_A$. Its off-diagonal entries count the records shared by pairs of rounds, while the independent mechanism noise contributes only $T\nu_\mu I_T$. The schedule therefore affects linear reconstruction through these overlaps.

3. RECONSTRUCTION AND RECORD REUSE

We compare schedules by the smallest variance achievable by an unbiased linear estimator, given by generalized least squares [16, 17]. This comparison requires only Assumption 1. When the non-target records are Gaussian, the same variance determines the Fisher information [18, 19] and also orders the optimal reconstruction risk for any loss.

3.1. Linear Reconstruction and Gaussian Reduction

Under Assumption 1, $w^\top Y$ is unbiased for x whenever $w^\top \mathbf{1}_T = 1$. Its variance is

$$r_x \|A^\top w\|_2^2 + T\nu_\mu \|w\|_2^2.$$

The first term is due to the unknown records, and the second to the mechanism noise. Uniform weights minimize the second term but need not minimize the total variance. Generalized least squares gives

$$v_A := \min_{w^\top \mathbf{1}_T = 1} w^\top \Sigma_A w = (\mathbf{1}_T^\top \Sigma_A^{-1} \mathbf{1}_T)^{-1}, \quad (3)$$

attained at $w_A = v_A \Sigma_A^{-1} \mathbf{1}_T$. To compare reconstruction beyond linear estimators, we make one additional assumption.

Assumption 2. Assumption 1 holds, and every centered non-target value is Gaussian.

Assumption 2 is used only for the information comparisons below. The privacy guarantees are calibrated on the bounded input domain and do not rely on this Gaussian population model. For a common target prior and nonnegative loss ℓ , define

$$\mathcal{R}_\ell(O) := \inf_{\delta} \mathbb{E}[\ell(X_\star, \delta(O))],$$

where the infimum is over measurable, possibly randomized rules with a common action space, and we assume the optimal risks are finite. An observation O_1 *Blackwell-dominates* O_2 if O_2 can be simulated from O_1 by an x -independent randomization [20]. Hence every attack based on O_2 can also be implemented from O_1 , so $\mathcal{R}_\ell(O_1) \leq \mathcal{R}_\ell(O_2)$. If each observation dominates the other, they are Blackwell equivalent.

Lemma 1. *Under Assumption 2, the statistic $Z_A := w_A^\top Y$ satisfies $Z_A | x \sim \mathcal{N}(x, v_A)$ and is Blackwell equivalent to Y . If $v_A \leq v_s$, the transcript dominates the source, so $\mathcal{R}_\ell(Y) \leq \mathcal{R}_\ell(Z_s)$; if $v_A \geq v_s$, the reverse dominance holds. They are equivalent at equality, and the dominance is strict for unequal variances if $X_\star$ can take two distinct values.*

Proof. The identities $w_A^\top \mathbf{1}_T = 1$ and $w_A^\top \Sigma_A w_A = v_A$ give the conditional law of Z_A . The residual $Y - \mathbf{1}_T Z_A$, conditional on x , has mean zero, covariance $\Sigma_A - v_A \mathbf{1}_T \mathbf{1}_T^\top$, and zero covariance with Z_A . Because the pair is jointly Gaussian, the residual is independent of Z_A , and its distribution does not depend on x . Thus Y can be generated from Z_A by adding an independent draw of this residual. Likewise, adding independent Gaussian noise converts the smaller-variance scalar observation into the larger-variance one. For unequal variances, equal-prior testing between two distinct target values has strictly larger Bayes error under the larger variance, so the reverse simulation is impossible. $\square$

By Lemma 1, under Assumption 2 the transcript is equivalent to observing x in Gaussian noise with variance v_A , and hence has Fisher information $\mathcal{I}(A) = 1/v_A$. Thus the variance comparison extends to all reconstruction rules.

3.2. Grouping at Fixed Reuse Counts

To separate how often records are used from how they are grouped across rounds, collect the reuse counts in $d := A^\top \mathbf{1}_T$. The transcript average is $\bar{Y} = x + d^\top u/T + \bar{\eta}$, where $\bar{\eta} \sim \mathcal{N}(0, \nu_\mu)$. Fixing d and the joint law of u fixes the distribution of $\bar{Y}$, whose variance is $v_{\text{avg}} = \nu_\mu + r_x \|d\|_2^2/T^2$. Thus, if two schedules have the same d , any difference in reconstruction must come from information lost by averaging.

For $T \geq 2$, let $D = [I_{T-1}, -\mathbf{1}_{T-1}]$. The differences DY remove the target but can still help estimate the error in the average, since $\text{Cov}(DY, \bar{Y} | x) = (r_x/T) DAd$. Every unbiased linear estimate can be written as $\bar{Y} - h^\top DY$. Minimizing its variance gives

$$Z_A = \bar{Y} - (r_x/T)(DAd)^\top (D\Sigma_A D^\top)^{-1} DY, \quad (4)$$

$$v_{\text{avg}} - v_A = (r_x^2/T^2)(DAd)^\top (D\Sigma_A D^\top)^{-1} DAd. \quad (5)$$

Averaging is suboptimal exactly when $DAd \neq 0$. The improvement decreases as ν_μ increases because $(D\Sigma_A D^\top)^{-1}$ decreases in the positive-semidefinite order.

The quantity $(Ad)_t$ sums the reuse counts of non-target records in round t . A schedule is *round-balanced* if this sum is constant across rounds, equivalently, $Ad = (\|d\|_2^2/T) \mathbf{1}_T$. For $T \geq 2$, this is equivalent to $DAd = 0$. It is also equivalent to $\Sigma_A \mathbf{1}_T \propto \mathbf{1}_T$, the classical condition under which ordinary averaging and generalized least squares agree [17]. Here, fixing d also fixes every record's privacy parameter and v_{avg} , so Theorem 1 isolates the effect of grouping alone.

Theorem 1. *Suppose Assumption 1 holds. Every binary participation matrix A , with $d = A^\top \mathbf{1}_T$, satisfies*

$$v_A \leq v_{\text{avg}} = \nu_\mu + r_x \|d\|_2^2/T^2, \quad (6)$$

with equality if and only if the schedule is round-balanced. Consequently, at fixed reuse counts d , every feasible round-balanced schedule maximizes v_A . Under Assumption 2, these

balanced transcripts are Blackwell equivalent, and every unbalanced transcript with the same d strictly dominates them if $X_\star$ can take two distinct values.

Proof. For $T \geq 2$, fixing d fixes v_{avg} . Equation (5) gives $v_A \leq v_{\text{avg}}$, with equality exactly when $DAd = 0$. For $T = 1$, every feasible schedule is balanced. Lemma 1 gives the Gaussian comparison. $\square$

For a balanced schedule, the round differences are uncorrelated with the error in the transcript average, so no unbiased linear estimator improves on averaging. Under Assumption 2, they are independent of the average error and provide no additional information. Without Gaussianity, nonlinear dependence may remain. If all records have the same reuse count, every schedule is round-balanced; hence, for linear estimation, grouping can matter only when reuse counts differ.

3.3. Persistence and the One-Time Benchmark

Unlike regrouping records with fixed reuse counts, changing how long records persist across rounds changes the uncertainty that remains in the transcript average. Equating the source variance with the optimal transcript variance gives

$$n^*(A) = 1 + (v_A - \nu_\mu)/r_x.$$

Under Assumption 2, the transcript dominates when $n > n^*(A)$, the source dominates when $n < n^*(A)$, and the two are equivalent at equality.

Suppose $m \in \{0, \dots, B-1\}$ non-target records appear in every round. Each round also contains $B-1-m$ fresh records that appear in no other round. Then

$$G_A = m \mathbf{1}_T \mathbf{1}_T^\top + (B-1-m) I_T, \\ (Ad)_t = mT + (B-1-m),$$

which does not depend on t . The schedule is therefore round-balanced, so averaging is optimal among unbiased linear estimators under Assumption 1. Its variance and the corresponding threshold are

$$v(m) = \nu_\mu + r_x (m + (B-1-m)/T), \quad (7)$$

$$n^*(m) = 1 + m + (B-1-m)/T. \quad (8)$$

Averaging reduces the variance contributed by fresh records by a factor of T , while the contribution from records that appear in every round remains unchanged. The mechanism-noise term is the same on both sides of the comparison and cancels from the threshold, so the ordering does not depend on μ .

Proposition 1. *Suppose the source and every transcript round use the same $n = B$ records. Under the independent mechanism noise in (2), the source and transcript are Blackwell equivalent for any joint distribution of these records.*

Proof. Let U be the common centered non-target sum. Then $Y_t = x + U + \eta_t$ and $\bar{Y} = x + U + \bar{\eta}$, where $\bar{\eta} \sim \mathcal{N}(0, \nu_\mu)$. The residual $Y - \mathbf{1}_T \bar{Y}$ contains only Gaussian mechanism noise. It is independent of $\bar{Y}$, and its distribution does not depend on x . Hence Y is equivalent to $\bar{Y}$. The source $Z_s = x + U + \eta_s$ has the same conditional distribution as $\bar{Y}$. $\square$

Here the round differences cancel both the target and the shared non-target sum. Thus, under the same total privacy calibration, repeatedly releasing the same aggregate reveals no more than a single release, regardless of the record distribution. This conclusion does not apply when the per-round noise is held fixed as more releases are added, since the overall privacy guarantee then becomes weaker.

3.4. Partial Target Participation

The conclusion changes when the target does not appear in every round. Differences between rounds with and without the target cancel the shared non-target sum but retain the target. As a result, target-free rounds can improve linear reconstruction even though the target has a stronger privacy guarantee.

Let $c \in \{0, 1\}^T$ be the public, data-independent target participation pattern, with $k = c^\top \mathbf{1}_T \geq 1$. Each round contains $B-1$ non-target records, and an absent target contributes zero. Keeping the per-round noise unchanged gives $Y = xc + Au + \eta$. The target's GDP parameter becomes $\mu\sqrt{k/T} \leq \mu$, while a record used in all rounds retains parameter μ ; replacing $X_\star$ leaves target-free rounds unchanged.

Proposition 2. *Suppose every round uses the same $B-1$ non-target records and $1 \leq k < T$. Let $\bar{Y}_j$ be the average over rounds with $c_t = j$. For any joint record distribution, the statistic $Z_{\text{diff}} := \bar{Y}_1 - \bar{Y}_0$ satisfies*

$$Z_{\text{diff}} = X_\star + \zeta, \quad \zeta \sim \mathcal{N}(0, T^2 \nu_\mu / [k(T-k)]). \quad (9)$$

The noise ζ is independent of all records. Moreover, Y is Blackwell equivalent to $(\bar{Y}_1, \bar{Y}_0)$. For any common prior and nonnegative loss with finite risks, $\mathcal{R}_\ell(Y) \leq \mathcal{R}_\ell(Z_{\text{diff}})$.

Proof. The common non-target sum cancels, leaving $X_\star + \bar{\eta}_1 - \bar{\eta}_0$. The two noise averages are independent, with total variance $T\nu_\mu/k + T\nu_\mu/(T-k)$. Within each participation group, subtracting the group mean leaves only Gaussian mechanism noise, independent of both group means and all records. Sampling these residuals reconstructs Y from the two means without using x , which proves equivalence. Since Z_{diff} is a function of Y , every attack based on Z_{diff} can also be implemented from Y . $\square$

The difference removes the shared records exactly but uses noise from both groups. The optimal linear estimator trades off these two terms. Under Assumption 1, it is $Z_c := v_A(c)c^\top \Sigma_A^{-1}Y$, where $v_A(c) := (c^\top \Sigma_A^{-1}c)^{-1}$. Under Assumption 2, Z_c is Blackwell equivalent to Y . For

persistent-fresh batches, write $a = T\nu_\mu + r_x(B-1-m)$ and $b = r_x m$. Applying the Sherman-Morrison formula to $\Sigma_A = aI_T + b\mathbf{1}_T\mathbf{1}_T^\top$ gives

$$v(m, k) = \frac{a(a+bT)}{k(a+b(T-k))}. \quad (10)$$

Corollary 1. *Suppose Assumption 1 holds, $m = B-1$, $n = B$, and $1 \leq k < T$. Then*

$$v(B-1, k) < v_s \iff kr_x(B-1) > T\nu_\mu. \quad (11)$$

Under Assumption 2, this condition characterizes strict transcript dominance if $X_\star$ can take two distinct values. For fixed T, k, ν_μ , the variance $v(B-1, k)$ increases from $T\nu_\mu/k$ to $T^2\nu_\mu/[k(T-k)]$ as $r_x(B-1)$ grows from 0 to ∞ .

Proof. Set $q = r_x(B-1)$ and $s = T\nu_\mu$. Equation (10) gives

$$\frac{v(B-1, k)}{v_s} = \frac{Ts}{k[s + (T-k)q]}.$$

Since $k < T$, this ratio is below one exactly when $kq > s$. Moreover, $\partial_q v = s^2/[s + (T-k)q]^2 > 0$, and the stated limits follow directly. Lemma 1 gives the Gaussian information ordering. $\square$

Condition (11) reads $(B-1)r_x > T\nu_\mu/k$: target-free rounds help once the variance of the shared non-target sum exceeds the noise variance in $\bar{Y}_1$. The lower limit $T\nu_\mu/k$, reached when the other records are known, exceeds ν_μ , as the target's guarantee requires. Restoring the target's parameter μ with less noise would weaken privacy for persistent records.

4. NUMERICAL STUDY

Figure 1 uses bounded records satisfying Assumption 1 but not Assumption 2. We draw independent records as $X = 2V - 1$, where V follows Beta(2, 6) with probability 0.7 and Beta(6, 2) otherwise. Thus $B_x = 1$ and $r_x = 22/75$.

We compare test mean-squared error (MSE) from histogram gradient boosting [21, 22] with the best affine MSE, $R_{\text{aff}}(v) = (r_x^{-1} + v^{-1})^{-1}$, where v is the minimum unbiased linear variance. The learned MSE is an upper bound on the Bayes risk but need not attain it. Each setting uses 10^5 training samples and 3×10^4 test samples per seed. Persistence and participation use $B = 16$, $T = 20$, and $\mu = 4$.

Grouping at fixed reuse counts. Table 1 compares two schedules with the same reuse counts ($T = 6$, $B = 4$). The paired schedules use the same records and mechanism noise, so their transcript averages are identical. Mean-only attacks observe $\bar{Y}$, while full-transcript attacks observe (Z_A, DY) .

Figure 1(a) shows lower MSE for the unbalanced transcript, with a larger gap as the mechanism noise decreases, as predicted by (5). At low noise, learned attacks also outperform mean-only attacks for the balanced schedule even

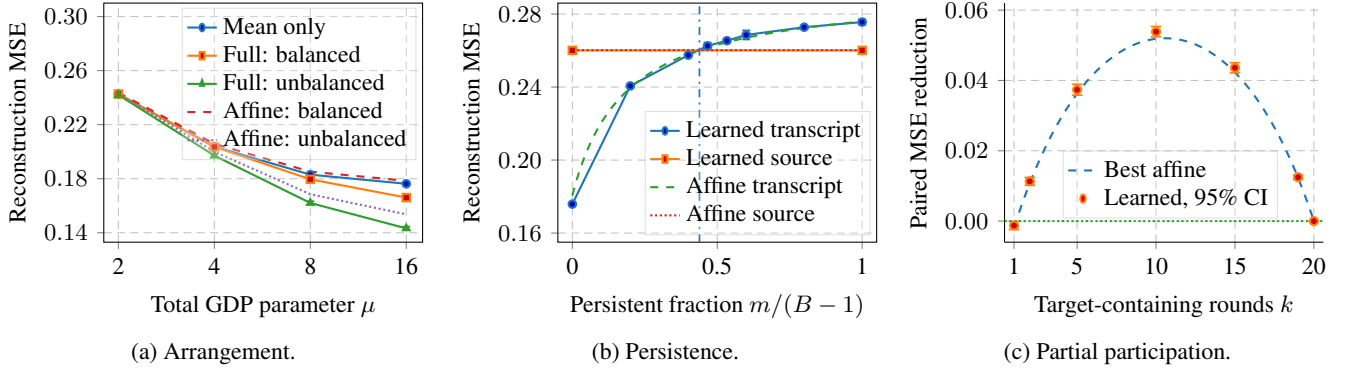


Fig. 1. Reconstruction on bounded records. (a,b) Learned MSEs (solid; one-standard-deviation bars over three seeds) and affine risks (dashed/dotted). The vertical line in (b) marks $m^*/(B-1) \approx 0.439$. (c) Paired MSE reduction, with 95% t intervals over five seeds; positive values favor the transcript. The target parameter is $4\sqrt{k/20}$, while persistent records retain parameter 4.

Table 1. Identical reuse counts, with per-round totals $Ad = 91_6$ (balanced) and $Ad = (9, 12, 3, 6, 12, 12)^T$ (unbalanced).

Round	1	2	3	4	5	6
Balanced	{1, 2, 4}	{1, 2, 5}	{1, 3, 6}	{1, 3, 7}	{2, 3, 8}	{2, 3, 9}
Unbalanced	{1, 2, 4}	{1, 2, 3}	{5, 6, 7}	{3, 8, 9}	{1, 2, 3}	{1, 2, 3}

though their affine risks are equal. Thus the round differences contain nonlinear information that Theorem 1 does not rule out.

Persistent and fresh contributions. We vary the number of persistent records while keeping the source size fixed at $n = 8$. Figure 1(b) shows the transcript MSE rising above the source MSE as more records persist across rounds. The affine comparison predicts a crossover at $m^* \approx 6.58$, where $n^*(m) = n$ in (8) (vertical line), and the learned curves cross nearby.

Partial target participation. Finally, with $n = B$, we vary target participation while reusing the same non-target batch and keeping the per-round noise fixed. For $k < T$, learned attacks observe $(Z_c, \bar{Y}_0)$, which determines $(\bar{Y}_1, \bar{Y}_0)$ and hence carries all information in Y . At $k = T$, we set $\eta_s = \bar{\eta}$, so the source and transcript average coincide. In Figure 1(c), fewer target appearances improve reconstruction despite the stronger target guarantee. Condition (11) predicts this for $2 \leq k < T$ but not for $k = 1$, where both reductions are slightly negative.

5. CONCLUSION

Record grouping can change reconstruction accuracy even when the record-wise privacy calibration and the distribution of the transcript average are fixed, because the same unknown non-target records can appear across several rounds. Under the same total privacy calibration, repeatedly releasing the same aggregate adds no information. Differences arise when

records are grouped unevenly, persist across rounds, or when some rounds omit the target. They vanish under strong privacy, where every attack approaches the prior, and grow as the shared records’ variance dominates the mechanism noise. Future work will study batch gradients in differentially private stochastic gradient descent (DP-SGD) [23], where queries change during training.

6. COMPLIANCE WITH ETHICAL STANDARDS

This is a numerical simulation study for which no ethical approval was required. The authors declare no conflicts of interest.

7. ACKNOWLEDGMENTS

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